

L^AT_EX Maths & Equations Practice Pack

10 Graded Exercises Modelled on the Lab-Exam Papers

Reproduce Part I from scratch — check Part II afterwards

July 21, 2026

Rules of engagement. Print Part I only. Start from a blank file with your own preamble. Time each attempt. An equation “passes” only if it *compiles* and matches structurally — the alignment column in the right place, the right lines numbered, delimiters grown to the right size, indices attached to the right symbols.

Contents

PART I — Target Equations	2
1 Reproduce each of the following	2
PART II — Solutions and Marking Notes	7
2 Do not read until you have attempted Part I	7
3 Drill Routine	13
4 Choosing the Right Environment	13
5 The Eight Errors That Cost the Most Marks	14

PART I — Target Equations

1 Reproduce each of the following

Exercise 1 — Warm-up

Difficulty: ★ **Target time:** 4 minutes

Tests: inline vs display maths, `equation` with `label`, `eqref` cross-reference, unnumbered display

Let $f(t)$ be absolutely integrable on $\mathbb{R}$. Its Fourier transform is defined by

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt, \quad \omega \in \mathbb{R}. \quad (1)$$

The inverse relation carries $F(\omega)$ back to the time domain:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega.$$

Equation (1) is the analysis equation, and the unnumbered display above is the synthesis equation.

Exercise 2 — Fractions, roots and big operators

Difficulty: ★★ **Target time:** 6 minutes

Tests: `dfrac`, `sqrt` with an index, `displaystyle`, limits above and below operators, `binom`

The three standard limits are

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e, \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0.$$

A closed form for the sum of squares is

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad n \in \mathbb{N}. \quad (2)$$

The binomial coefficient and a nested radical are

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \sqrt[3]{\frac{a^2 + b^2}{c}}, \quad \prod_{i=1}^n \frac{1}{1-x_i}.$$

Inline, the same sum is written $\sum_{k=1}^n k^2$, which is deliberately smaller than its displayed form.

Exercise 3 — A derivation chain, only the last line numbered

Difficulty: ★★★ **Target time:** 8 minutes

Tests: `align`, the `&` alignment column, `nonumber` on intermediate steps

Applying the Laplace transform to a series RL circuit with $i(0) = 0$ gives

$$\begin{aligned} L[sI(s) - i(0)] + RI(s) &= V(s), \\ \Rightarrow sLI(s) + RI(s) &= V(s), \\ \Rightarrow I(s)(sL + R) &= V(s). \end{aligned} \quad (3)$$

Rearranging Equation (3) produces the transfer function

$$H(s) = \frac{I(s)}{V(s)} = \frac{1}{sL + R} = \frac{1}{L} \cdot \frac{1}{s + R/L}. \quad (4)$$

Exercise 4 — Several lines, one number

Difficulty: ★★★ **Target time:** 6 minutes

Tests: aligned nested inside equation, versus align; gathered

For a second-order system the natural frequency and damping ratio are

$$\begin{aligned}\omega_n &= \frac{1}{\sqrt{LC}} \\ \zeta &= \frac{R}{2} \sqrt{\frac{C}{L}}\end{aligned}\tag{5}$$

whereas the same two facts written without an alignment column are

$$\begin{aligned}\omega_n^2 &= \frac{1}{LC} \\ 2\zeta\omega_n &= \frac{R}{L}\end{aligned}\tag{6}$$

Exercise 5 — Piecewise definitions and matrices

Difficulty: *** **Target time:** 8 minutes

Tests: cases, bmatrix, pmatrix, vmatrix, text inside maths

The unit-step and signum functions are

$$u(t-a) = \begin{cases} 0 & \text{if } t < a, \\ 1 & \text{if } t \geq a, \end{cases} \quad \text{sgn}(x) = \begin{cases} -1 & x < 0, \\ 0 & x = 0, \\ 1 & x > 0. \end{cases}$$

For the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}, \quad (7)$$

the eigenvalue problem reads

$$\begin{pmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Exercise 6 — Growing delimiters and evaluation bars

Difficulty: *** **Target time:** 8 minutes

Tests: left/right auto-sizing, limits attached to a closing bracket, integration by parts as a chain

Integrating by parts,

$$\begin{aligned} \mathcal{L}\{t\} &= \int_0^\infty t e^{-st} dt = \left[-\frac{t}{s} e^{-st} \right]_0^\infty + \frac{1}{s} \int_0^\infty e^{-st} dt \\ &= 0 + \frac{1}{s} \left[-\frac{1}{s} e^{-st} \right]_0^\infty \\ &= \frac{1}{s^2}, \quad s > 0. \end{aligned} \quad (8)$$

Delimiters must grow with their contents:

$$\left(\frac{a}{b}\right), \quad \left[\frac{a}{b}\right], \quad \left\{\frac{a}{b}\right\}, \quad \left|\frac{a}{b}\right|, \quad \left\langle\frac{a}{b}\right\rangle, \quad \left\lfloor\frac{a}{b}\right\rfloor.$$

A one-sided delimiter is also possible:

$$\left. \frac{dv_C}{dt} \right|_{t=0} = \frac{V_0}{RC}.$$

Exercise 7 — Dense indices and tensor notation

Difficulty: ★★★ **Target time:** 12 minutes

Tests: stacked sub/superscripts on one symbol, `\varphi` vs `\phi`, negative thin space, upright differentials

Einstein's field equations are

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (9)$$

For a spherically symmetric mass M the line element is

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2\theta d\varphi^2), \quad r_s = \frac{2GM}{c^2}. \quad (10)$$

A freely falling particle satisfies the geodesic equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (11)$$

and the perihelion advance per orbit is

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}. \quad (12)$$

Index gymnastics to reproduce exactly:

$$A^\mu{}_\nu, \quad R^\rho{}_{\sigma\mu\nu}, \quad {}^a_b X^c_d, \quad T^{\mu\nu}{}_{;\lambda}, \quad x_{i_{jk}}, \quad e^{-t/(2\tau)}.$$

Exercise 8 — The other alignment environments

Difficulty: ★★★ **Target time:** 10 minutes

Tests: `gather`, `multline`, `alignat`, `subequations`

Independent centred equations with `gather`:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \quad (13)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (14)$$

One long formula broken with `multline`:

$$(a + b + c + d)^2 = a^2 + b^2 + c^2 + d^2 + 2ab + 2ac + 2ad + 2bc + 2bd + 2cd \quad (15)$$

Two aligned pairs on one line with `alignat`:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad (16)$$

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan \frac{y}{x}. \quad (17)$$

Sub-numbered equations with `subequations`:

$$\nabla \cdot \mathbf{B} = 0 \quad (18a)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (18b)$$

Refer to the pair as (18) and to one member as (18b).

Exercise 9 — Theorem, proof, custom operators, forced tags

Difficulty: ★★★ **Target time:** 10 minutes
Tests: amsthm environments, tag, DeclareMathOperator, substack

Definition 1.1. A signal f is said to be of *exponential order* σ if there exist constants $M > 0$ and $T \geq 0$ such that $|f(t)| \leq Me^{\sigma t}$ for all $t > T$.

Proposition 1.1 (Weak-field circular orbit). *For a circular orbit of radius r about a mass M ,*

$$\left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2} = \frac{r_s}{2r}. \quad (*)$$

Proof. The circular-orbit force balance $v^2/r = GM/r^2$ gives $v^2 = GM/r$. Dividing by c^2 and substituting $r_s = 2GM/c^2$ yields $(*)$. $\square$

Operators declared in the preamble print upright and carry their limits:

$$\arg \min_{\substack{\theta \in \Theta \\ \theta \neq 0}} \sum_{i=1}^n (y_i - f(x_i; \theta))^2, \quad \mathcal{L}\{f(t)\} = F(s).$$

See Proposition 1.1 and Equation $(*)$.

Exercise 10 — Full exam simulation

Difficulty: ★★★★★ **Target time:** 15 minutes
Tests: a numbered derivation, text and units inside maths, a footnote, annotated braces, a boxed result and two cross-references

Consider the first-order system of Equation (4) driven by a constant input V_0 .¹ In the transform domain,

$$V(s) = \frac{V_0}{s} \quad \implies \quad I(s) = \frac{V_0}{s(sL + R)}. \quad (19)$$

Partial fractions give

$$\begin{aligned} I(s) &= \frac{V_0}{R} \left(\frac{1}{s} - \frac{1}{s + R/L} \right) \\ &= \underbrace{\frac{V_0}{R} \cdot \frac{1}{s}}_{\text{steady state}} - \underbrace{\frac{V_0}{R} \cdot \frac{1}{s + R/L}}_{\text{transient}}. \end{aligned} \quad (20)$$

Taking the inverse transform of Equation (20),

$$\boxed{i(t) = \frac{V_0}{R} \left(1 - e^{-Rt/L} \right), \quad t \geq 0}$$

where the time constant is $\tau = L/R$. For $L = 10$ mH and $R = 2$ Ω we obtain $\tau = 5$ ms, so the current reaches 63.2% of its final value V_0/R after 5 ms.

¹All quantities are in SI units unless stated otherwise.

PART II — Solutions and Marking Notes

2 Do not read until you have attempted Part I

Solution 1

Let $f(t)$ be absolutely integrable on $\mathbb{R}$. Its Fourier transform is defined by

```
\begin{equation}
F(\omega) = \int_{-\infty}^{\infty} f(t) \, e^{-j\omega t} \, dt,
\quad \omega \in \mathbb{R}.
\label{eq:ft}
\end{equation}
```

The inverse relation carries $F(\omega)$ back to the time domain:

```
\[
f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \, e^{j\omega t} \, d\omega.
\]
```

Equation~\eqref{eq:ft} is the analysis equation.

Marking notes. Three modes, three delimiters: $\dots$ for inline, $\left[\dots \right]$ for an unnumbered display, `equation` for a numbered display. Never use $\$ \$ \dots \$ \$$ — it is plain-T_EX syntax that produces wrong vertical spacing and breaks some packages. `\eqref` prints (1) with parentheses while `\ref` prints a bare 1; for equations always use `\eqref`. The `\,` before `dt` is a thin space, and `-\infty` needs no braces because it is a single token, but `e^{-j\omega t}` does.

Solution 2

```
\[
\lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right)^n = e,
\quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1,
\quad \lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0.
\]
```

```
\begin{equation}
\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6},
\quad n \in \mathbb{N}.
\label{eq:squares}
\end{equation}
```

```
\[
\binom{n}{k} = \frac{n!}{k! (n-k)!},
\quad \sqrt[3]{\frac{a^2 + b^2}{c}},
\quad \prod_{i=1}^n \frac{1}{1 - x_i}.
\]
```

Inline, the same sum is $\sum_{k=1}^n k^2$.

Marking notes. In a *display*, `\sum_{k=1}^n` automatically places its limits above and below. *Inline*, the same code moves them to the side and shrinks everything, to protect the line spacing — force the big form with $\displaystyle \sum_{k=1}^n$ if the target shows it. `\frac` adapts to context; `\dfrac` is always display-size and `\tfrac` always text-size. The root index is an *optional* argument: `\sqrt[3]{\dots}`. Function names must be commands — `\sin`, `\ln`, `\arctan` — or they print in italic as a product of variables, which loses marks.

Solution 3

```
\begin{align}
L\bigl[ sI(s) - i(0) \bigr] + R\,I(s) &= V(s), \nonumber \\
\Rightarrow sL\,I(s) + R\,I(s) &= V(s), \nonumber \\
\Rightarrow I(s)\,(sL + R) &= V(s). \\
\label{eq:rl}
\end{align}
\begin{equation}
H(s) = \frac{I(s)}{V(s)} = \frac{1}{sL + R} \\
= \frac{1}{L}\cdot\frac{1}{s + R/L}. \\
\label{eq:rltf}
\end{equation}
```

Marking notes. This is the single most common exam pattern. Put the *&* immediately before the = on every line: everything left of *&* is right-aligned, everything right of it is left-aligned, so all the equals signs stack into one column. `\nonumber` ends a line — never put one after the final line, or you get a spurious empty row.

`\nonumber` (identical to `\notag`) suppresses that line’s number. Here it goes on the first two lines so only the conclusion is numbered, exactly as printed. Two shortcuts worth knowing: `align*` suppresses *all* numbers, and `\label` may sit anywhere inside the environment but attaches to the line it appears on. Use `\bigl[ ... \bigr]` for brackets slightly larger than default but not as aggressive as `\left[ ... \right]`; the thin space `\,` in `R\,I(s)` stops two italic letters from colliding.

Solution 4

```
\begin{equation}
\begin{aligned}
\omega_n &= \frac{1}{\sqrt{LC}} \\
\zeta &= \frac{R}{2}\sqrt{\frac{C}{L}}
\end{aligned}
\label{eq:secondorder}
\end{equation}
\begin{equation}
\begin{gathered}
\omega_n^2 = \frac{1}{LC} \\
2\zeta\omega_n = \frac{R}{L}
\end{gathered}
\end{equation}
```

Marking notes. The distinction is worth memorising because the exam papers use both:

Want	Use
Several lines, each numbered	<code>align</code>
Several lines, ONE number	<code>equation + aligned</code>
Several lines, no alignment, each numbered	<code>gather</code>
Several lines, no alignment, ONE number	<code>equation + gathered</code>

The `-ed` suffix means “inner version”: `aligned` and `gathered` produce a *box* that carries no number of its own, so the enclosing `equation` supplies exactly one. Putting `align` inside `equation` is an error — it will refuse to compile.

Solution 5

```
\[
```



```

u(t-a) = \begin{cases}
0 & \text{if } t < a, \\
1 & \text{if } t \geq a,
\end{cases}
\quad
\operatorname{sgn}(x) = \begin{cases}
-1 & x < 0, \\
0 & x = 0, \\
1 & x > 0.
\end{cases}
\\
\begin{equation}
A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},
\quad
\det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
= a_{11}a_{22} - a_{12}a_{21},
\label{eq:det}
\end{equation}

```

Marking notes. In `cases` the `&` separates the *value* from the *condition*, and `\\` starts a new case — the same two characters as a table, doing an analogous job. Note the space in `\text{if }`: a trailing space inside `\text` is preserved, whereas maths mode would swallow it. Bracket style is chosen purely by the environment name: `bmatrix` for square, `pmatrix` for round, `vmatrix` for single bars (a determinant), `Vmatrix` for double bars, `Bmatrix` for braces, `matrix` for none. `\operatorname{sgn}` creates an upright one-off operator without a preamble declaration; if you need it repeatedly, declare it once instead.

Solution 6

```

\begin{align}
\operatorname{Lap}\{t\} &= \int_0^{\infty} t \, e^{-st} \, dt \\
&= \left[ -\frac{t}{s} e^{-st} \right]_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} \, dt \\
&= 0 + \frac{1}{s} \left[ -\frac{1}{s} e^{-st} \right]_0^{\infty} \\
&= \frac{1}{s^2}, \quad s > 0.
\label{eq:lapt}
\end{align}
[
\left. \frac{dV_C}{dt} \right|_{t=0} = \frac{V_0}{RC}.
]

```

Marking notes. `\left` and `\right` must appear as a **pair** in the same line of the same environment; an unmatched one gives Extra }, or forgotten `\right`. Because they pair, spanning them across a `\\` is illegal — if a bracket must open on one line and close on the next, use `\left...\right`. then `\left...\right` with invisible partners.

The evaluation limits attach to the closing bracket: `\right]_0^{\infty}`. For the one-sided vertical bar, `\left.` is an invisible opening delimiter, so `\left. ... \right|_{t=0}` gives a bar that grows with the fraction, and `\left| ... \right|` only groups and prints nothing — that is why `\operatorname{Lap}\{t\}` needs the backslash.

Solution 7

```

\begin{equation}
G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi}{c^4} T_{\mu\nu}.
\label{eq:efe}
\end{equation}
\begin{equation}

```

```

ds^{2} = -\left( 1 - \frac{r_{\text{s}}}{r} \right) c^{2} \backslash, dt^{2}
+ \left( 1 - \frac{r_{\text{s}}}{r} \right)^{-1} dr^{2}
+ r^{2} \left( d\theta^{2} + \sin^{2}\theta \, d\varphi^{2} \right),
\qquad r_{\text{s}} = \frac{2GM}{c^{2}} .
\label{eq:metric}
\end{equation}
\begin{equation}
\frac{d}{d\tau} x^{\mu} \frac{d}{d\tau} \tau
+ \Gamma^{\mu}_{\alpha\beta} \frac{d}{d\tau} \alpha \frac{d}{d\tau} \beta \backslash, \frac{dx^{\beta}}{d\tau} = 0,
\label{eq:geodesic}
\end{equation}
\begin{equation}
\Delta\varphi = \frac{6\pi G M}{a(1 - e^{2})} c^{2} .
\end{equation}
\[
A^{\mu}_{\nu}, \qquad R^{\rho}_{\sigma\mu\nu}, \qquad
{}^{\text{a}}{}_{{\text{b}}}X^{\text{c}}_{\text{d}}, \qquad T^{\mu\nu}_{\phantom{\mu\nu}}; \backslash, \lambda, \qquad
x_{i_j\{k\}}, \qquad e^{-t/(2\tau)} .
\]

```

Marking notes. Everything here reduces to one rule: **group with braces**. $\Gamma^{\mu}_{\alpha\beta}$ puts μ above and $\alpha\beta$ below the *same* Γ ; the order of $\wedge$ and $_$ does not matter, but omitting the braces does — $\Gamma^{\mu}_{\alpha\beta}$ drops β to the baseline.

To stop indices stacking vertically when they should be staggered, insert a phantom space: A^{μ}_{ν}. For a leading index, open with an empty group: $\{{}^{\text{a}}{}_{{\text{b}}}\text{X}$. The same trick separates two index groups on one symbol: $T^{\mu\nu}_{}; \backslash, \lambda$.

φ is the curly φ ; ϕ is the straight ϕ . Choosing the wrong one is a silent lost mark, and the same applies to ϵ vs ε and ρ vs ϱ . The $\!$ in $\sin^{2}\!\theta$ is a *negative* thin space that pulls θ back against the exponent. Finally, $\wedge^{-1}$ after $\right)$ raises the whole grown parenthesis to that power.

Solution 8

```

\begin{gather}
\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \backslash \backslash
\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J}
+ \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}
\end{gather}

\begin{multline}
(a + b + c + d)^2 = a^2 + b^2 + c^2 + d^2 \backslash \backslash
+ 2ab + 2ac + 2ad + 2bc + 2bd + 2cd
\end{multline}

\begin{alignat}{2}
x &= r \cos \theta, & \qquad y &= r \sin \theta, \backslash \backslash
r &= \sqrt{x^2 + y^2}, & \qquad \theta &= \arctan \frac{y}{x}.
\end{alignat}

\begin{subequations}
\label{eq:maxwell}
\begin{align}
\boldsymbol{\nabla} \cdot \mathbf{B} &= 0 \label{eq:maxb} \backslash \backslash
\boldsymbol{\nabla} \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \label{eq:maxe}
\end{align}
\end{subequations}

```

Marking notes. `gather` centres each line independently and takes **no** `&` — putting one there is an error. `multline` puts the first line flush left and the last flush right, with no alignment column; `\shoveleft` or `\shoveright` override an individual middle line.

`alignat{2}` takes a mandatory argument: the number of alignment *pairs*. It inserts *zero* space between the pairs, which is the whole point — you control the gap yourself with `\qqquad`. Plain `align` would insert its own stretchy space and push the pairs to the margins.

`subequations` wraps a group so it numbers (7a), (7b) instead of (7), (8). Put a `\label` directly after `\begin{subequations}` to reference the whole group, and individual labels inside for the members.

Solution 9

```
% ---- preamble ----
\usepackage{amsthm}
\DeclareMathOperator{\Lap}{\mathcal{L}}
\DeclareMathOperator*{\argmin}{arg\,min}
\theoremstyle{plain}
\newtheorem{theorem}{Theorem}[section]
\newtheorem{proposition}[theorem]{Proposition}
\theoremstyle{definition}
\newtheorem{definition}[theorem]{Definition}[section]

% ---- body ----
\begin{proposition}[Weak-field circular orbit]\label{prop:weak}
For a circular orbit of radius  $r$  about a mass  $M$ ,
\begin{equation}
\left( \frac{v}{c} \right)^2 = \frac{GM}{r c^2} = \frac{r_s}{2r}.
\tag{$\ast$}
\label{eq:weak}
\end{equation}
\end{proposition}

\begin{proof}
The circular-orbit force balance  $v^2/r = GM/r^2$  gives  $v^2 = GM/r$ .
Dividing by  $c^2$  and substituting  $r_s = 2GM/c^2$  yields~\eqref{eq:weak}.
\end{proof}

\[
\argmin_{\substack{\theta \in \Theta \\ \theta \neq 0}}
\; \sum_{i=1}^n \bigl( y_i - f(x_i; \theta) \bigr)^2
\]
```

Marking notes. `\newtheorem{proposition}[theorem]{Proposition}` — the *optional* `[theorem]` in the middle makes propositions share the theorem counter, while the trailing `[section]` on the first declaration ties numbering to the section, producing “Proposition 3.1”. The optional argument to the environment itself, `\begin{proposition}[Name]`, prints the parenthetical title.

`\tag{$\ast$}` overrides the automatic number with anything you like and does *not* advance the counter; `\tag*{...}` omits the parentheses. This is how you match a target whose numbering you cannot otherwise reproduce.

`\DeclareMathOperator` makes an upright multi-letter operator; the starred form `\DeclareMathOperator*` puts limits *below* in display style, which is why `\argmin` carries its condition underneath. `\substack{a \\\ b}` stacks a multi-line subscript. The `proof` environment supplies the italic lead-in and the closing $\square$ automatically; `\qedhere` moves that square onto the last display line when the proof ends with an equation.

Solution 10

```

Consider the first-order system of Equation~\eqref{eq:rltf} driven by a
constant input  $V_0$ .\footnote{All quantities are in SI units unless
stated otherwise.} In the transform domain,
\begin{equation}
V(s) = \frac{V_0}{s}
\quad \Longrightarrow \quad
I(s) = \frac{V_0}{s(sL + R)} .
\label{eq:drive}
\end{equation}
Partial fractions give
\begin{align}
I(s) &= \frac{V_0}{R}
\left( \frac{1}{s} - \frac{1}{s + R/L} \right) \nonumber \\
&= \underbrace{\frac{V_0}{R} \cdot \frac{1}{s}}_{\text{steady state}}
- \underbrace{\frac{V_0}{R} \cdot \frac{1}{s + R/L}}_{\text{transient}} .
\label{eq:pf}
\end{align}
\[
\boxed{\;
i(t) = \frac{V_0}{R} \left( 1 - e^{-Rt/L} \right),
\quad t \geq 0 \; }
\]
where the time constant is  $\tau = L/R$ . For  $L = 10$  mH and
 $R = 2$   $\Omega$  we obtain  $\tau = 5$  ms, so the current reaches
63.2% of its final value  $V_0/R$ .

```

Marking notes. Six graded features at once. `\underbrace{expr}_{\text{label}}` draws the brace *below* with the label beneath it; `\overbrace{expr}^{\text{label}}` does the mirror image. `\boxed{...}` frames a result, and the `\;` inside adds breathing room so the box does not clamp the maths. Units are upright text inside maths: `10 \text{mH}`, with `\` (backslash-space) as a normal inter-word space — writing `10 mH` bare would render as the product $10mH$ in italic. Greek letters used as units, like `\Omega`, are ordinary maths. The percent sign still needs escaping as `63.2\%` even in maths mode. Finally the `\footnote` sits in body text, so a plain `\footnote` works; inside a float you would need `\footnotemark/\footnotetext`.

3 Drill Routine

1. **Round 1 — untimed.** Exercises 1, 2, 4 with Part II open. The aim is to fix the environment-selection table from Solution 4 in memory.
2. **Round 2 — the core pattern.** Exercise 3 alone, five times in a row, until the `&/\nonumber` chain is automatic. This single pattern appeared in both exam papers and is worth the most marks per minute.
3. **Round 3 — timed, structural.** Exercises 5, 6, 8 in 25 minutes.
4. **Round 4 — the index gauntlet.** Exercise 7 in 12 minutes. Then rewrite only the final index line from memory until it is error-free.
5. **Round 5 — exam pace.** Exercises 9 and 10 back to back in 25 minutes, including preamble declarations and cross-references.
6. **Round 6 — from memory.** Reproduce Exercise 3 and Exercise 7 with nothing open. When those two are automatic, equations will not cost you marks.

4 Choosing the Right Environment

Situation	Environment
Symbol in a sentence	<code>\$...\$</code>
Standalone, no number	<code>\[...\]</code>
Standalone, numbered	<code>equation</code>
Multi-line, aligned, each numbered	<code>align</code>
Multi-line, aligned, no numbers	<code>align*</code>
Multi-line, aligned, ONE number	<code>equation + aligned</code>
Multi-line, centred, each numbered	<code>gather</code>
Multi-line, centred, ONE number	<code>equation + gathered</code>
One long formula split over lines	<code>multline</code>
Two or more aligned pairs per line	<code>alignat{n}</code>
Numbered (1a), (1b)	<code>subequations</code>
Piecewise definition	<code>cases</code>
Suppress one line's number	<code>\nonumber</code> or <code>\notag</code>
Force a specific number	<code>\tag {...}</code>

5 The Eight Errors That Cost the Most Marks

#	Mistake	Symptom and fix
1	<code>x^2^3</code> or <code>x_i_j</code>	Double superscript. Group them: <code>x^{2^{3}}</code> , <code>x_{i_{j}}</code> .
2	Unbraced multi-token index	<code>e^-st</code> prints e^{-st} . Always <code>e^{-st}</code> .
3	<code>\left</code> without <code>\right</code>	Extra <code>}</code> , or forgotten <code>\right</code> . They must pair within one line; use <code>\right .</code> for an invisible partner.
4	Maths symbol left in text	Missing <code>\$</code> inserted. Wrap it: <code>\$_alpha\$</code> .
5	Words italic inside maths	<code>if</code> renders as $i\cdot f$. Use <code>\text{if}</code> with the space <i>inside</i> the braces.
6	<code>&</code> used in <code>gather</code>	Misplaced alignment tab. <code>gather</code> takes no alignment column — switch to <code>align</code> .
7	<code>\\</code> after the final line	A blank numbered row appears at the end. Delete it.
8	Compiling once	<code>\eqref</code> shows <code>??</code> . Run <code>pdflatex</code> twice.

Put the $\&$ before the `=`, and `\nonumber` on every line but the last.